Assignment 9

1. Application, Presentation, Session Layers

These three layers work together to provide network services to end users:

- Application Layer: Provides services directly to user applications (e.g., web browsers, email clients).
- Presentation Layer: Translates, encrypts, and compresses data to ensure proper format between systems.
- Session Layer: Establishes, manages, and terminates communication sessions between devices.

Together, they ensure data is correctly formatted, securely transmitted, and properly managed during communication.

2. Web and Email Protocols

Web protocols:
- HTTP/HTTPS: Used for transferring web pages between client and server.

Email protocols:
- SMTP: Used to send emails.
- POP3: Retrieves emails and downloads them locally.
- IMAP: Retrieves emails while keeping them on the server.

These protocols allow communication over the internet for browsing and messaging.

3. DNS and DHCP

- DNS (Domain Name System): Translates domain names (like google.com) into IP addresses.
- DHCP (Dynamic Host Configuration Protocol): Automatically assigns IP addresses and network settings to devices.

DNS helps locate servers, while DHCP helps devices join the network.

4. File Transfer Protocols
File transfer protocols like FTP and SFTP are used to transfer files between a client and server.

They work by:
- Establishing a connection.
- Authenticating the user.
- Transferring files using commands (upload/download).
- Closing the connection after completion.

5. POP vs IMAP
POP:
- Downloads emails to local device.
- Usually deletes from server.
- Works offline.

IMAP:
- Keeps emails on server.
- Syncs across multiple devices.
- Requires internet connection.

IMAP is more flexible, POP is simpler.

6. DHCP Messages
DHCP uses four main messages (DORA process):
- Discover: Client searches for DHCP server.
- Offer: Server offers IP address.
- Request: Client requests offered IP.
- Acknowledge: Server confirms assignment.

7. FTP Client Operation
An FTP client connects to an FTP server using login credentials.

Steps:
- Establish connection.
- Authenticate user.
- Browse server files.

- Upload (push) or download (pull) files.
- End session.

Data is transferred using control and data connections.

8. SMB Protocol

SMB (Server Message Block) is used for file and resource sharing over networks.

Functions of SMB messages:
- File access (read/write files).
- Resource sharing (printers, folders).
- Communication between client and server.

9. FTP vs SMB

FTP:
- Used for file transfer.
- Works over internet.
- Requires separate client.

SMB:
- Used for file sharing within networks.
- Common in LAN environments.
- Integrated with operating systems.

FTP is for transfer; SMB is for sharing.

10. SMTP vs POP vs IMAP

SMTP:
- Sends emails.

POP:
- Downloads emails.

IMAP:
- Synchronizes emails across devices.

SMTP is for sending, POP and IMAP are for receiving, with IMAP offering better synchronization.